DIGITAL LOGIC — QUICK HEURISTICS

Q1 · K-MAP

- `1s → SOP`
- `0s → POS`
- big groups first
- unchanged vars survive

Q2 · DON'T CARE

- `X = use if helpful`

Q3 · FULL ADDER

- `S = x ⊕ y ⊕ z`
- `C = xy + xz + yz`

Q4 · ADD / SUB

- `M=0 → A+B`
- `M=1 → A−B`
- `Y = A ⊕ M`
- `Cin = M`

Q5 · BCD

- ADD
- `→ correction?`
- `K + Z8Z4 + Z8Z2`
- `→ +0110`

Q6 · MUX

- truth table
- `→ pairs`
- `→ 0 / 1 / C / C′`

Q7 · DECODER

- minterms
- `→ decoder`
- `→ OR outputs`

Q8 · COMPARATOR

- `MSB → LSB`
- equality bits

Q9 · PRIORITY ENCODER

- highest input wins
- `V = OR(inputs)`

Q10 · FF EXCITATION

- `00 → J0 KX / T0`
- `01 → J1 KX / T1`
- `10 → JX K1 / T1`
- `11 → JX K0 / T0`

Q11 · SEQUENTIAL DESIGN

- `diagram → table → assignment → excitation`
- `→ K-maps → equations → circuit`

Q12 · MOD COUNTER

- `valid states → excitation → K-maps`
- `→ equations → unused-state check`

Q13 · ANALYSIS

- `equations → calculate next states`
- `→ state table → state diagram`

Q14 · COUNTERS

- `RING = n`
- `JOHNSON = 2n`

EXAM MOVE: Identify the question type → apply the matching arrow-chain → check the final state/output.